Sample Size Computations

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```r
library(pwr)
```

```
## Warning: package 'pwr' was built under R version 4.5.2
```

```r
find_sample_size <- function(means, sigma2, alpha = 0.05, target_power = 0.90) {

  a <- length(means)


  mu_bar <- mean(means)


  tau <- means - mu_bar


  sum_tau2 <- sum(tau^2)

  max_diff <- max(means) - min(means)

  cat("=== Input Summary ===\n")
  cat("Number of groups (a):           ", a, "\n")
  cat("Group means:                    ", means, "\n")
  cat("Overall mean:                   ", mu_bar, "\n")
  cat("Treatment effects (tau_i):      ", tau, "\n")
  cat("Sum of squared effects:         ", sum_tau2, "\n")
  cat("Maximum difference:             ", max_diff, "\n")
  cat("Error variance (sigma^2):       ", sigma2, "\n")
  cat("Alpha:                          ", alpha, "\n")
  cat("Target power:                   ", target_power, "\n")
  cat("\n=== Searching for n ===\n")


for (n in 2:200) {


    phi2 <- (n * sum_tau2) / (a * sigma2)
    phi  <- sqrt(phi2)


    df1 <- a - 1
    df2 <- a * (n - 1)
```

```r
    N   <- a * n


    F_crit <- qf(1 - alpha, df1, df2)

    ncp   <- (n * sum_tau2) / sigma2
    power <- pf(F_crit, df1, df2, ncp = ncp, lower.tail = FALSE)

    cat(sprintf("n = %3d | Phi^2 = %6.3f | Phi = %5.3f | df2 = %3d | Power = %.4f\n",
                n, phi2, phi, df2, power))

    if (power >= target_power) {
      cat("\n=== Result ===\n")
      cat("Required sample size per group: n =", n, "\n")
      cat("Total observations needed:      N =", N, "\n")
      cat("Achieved power:                    ", round(power, 4), "\n")
      cat("Phi^2:                             ", round(phi2, 4), "\n")
      cat("Phi:                               ", round(phi,  4), "\n")
      return(invisible(list(n = n, N = N, power = power,
                            phi2 = phi2, phi = phi,
                            df1 = df1, df2 = df2)))
    }
  }
  cat("No Solution for this power level")
}
means         <- c(50, 60, 50, 60)
sigma2        <- 25
alpha         <- 0.05
target_power  <- 0.90

result <- find_sample_size(means, sigma2, alpha, target_power)
```

```
## === Input Summary ===
## Number of groups (a):          4
## Group means:                   50 60 50 60
## Overall mean:                  55
## Treatment effects (tau_i):     -5 5 -5 5
## Sum of squared effects:        100
## Maximum difference:            10
## Error variance (sigma^2):      25
## Alpha:                         0.05
## Target power:                  0.9
##
## === Searching for n ===
## n =   2 | Phi^2 =  2.000 | Phi = 1.414 | df2 =   4 | Power = 0.2987
## n =   3 | Phi^2 =  3.000 | Phi = 1.732 | df2 =   8 | Power = 0.6157
## n =   4 | Phi^2 =  4.000 | Phi = 2.000 | df2 =  12 | Power = 0.8224
## n =   5 | Phi^2 =  5.000 | Phi = 2.236 | df2 =  16 | Power = 0.9270
##
## === Result ===
## Required sample size per group: n = 5
## Total observations needed:      N = 20
## Achieved power:                     0.927
```

```
## Phi^2:                                5
## Phi:                                  2.2361
```